

Course: Internet of Things and Applied Data Science (Fall 2025)

Project Title: Smart Home Temperature Prediction using IoT Sensor Data

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1. Introduction & Problem Statement

Problem Definition In this project, we address the efficiency of HVAC systems in smart homes. Using the Problem Statement Template provided in the course guidelines:

- **We want to:** Predict future temperature for Smart Home HVAC Systems.
- **Because it impacts:** Energy efficiency, heating costs, and resident comfort.
- **Using data from:** IoT Weather/Temperature Sensors at 10-minute intervals.
- **Success looks like:** Predicting temperature with a Mean Absolute Error (MAE) of less than 1°C.
- **Constraints:** Compute power (AWS Free Tier limitations), Latency.

Scope

- **In Scope:** Data ingestion from CSV, data cleaning (handling missing values), baseline visualization, and deploying a Linear Regression model.
- **Out of Scope:** Real-time hardware integration and custom sensor development.

2. Data Source

We used the Jena Climate Dataset (or similar "Smart Home Energy/Weather" dataset), which is a public time-series dataset sourced from Kaggle (as suggested in Project Guidelines Section 4).

- **Dataset Content:** The dataset contains 10-minute interval recordings of weather parameters such as temperature, humidity, and pressure.
- **Preprocessing:** We cleaned the data by dropping NaN values and formatted the timestamp column for time-series analysis.

3. Methodology

Tools Used

- **Python:** For data manipulation and modeling.
- **Pandas & NumPy:** For data cleaning.
- **Matplotlib:** For visualization.
- **Scikit-Learn:** For implementing the Machine Learning model.

- **GitHub:** For version control and collaboration.

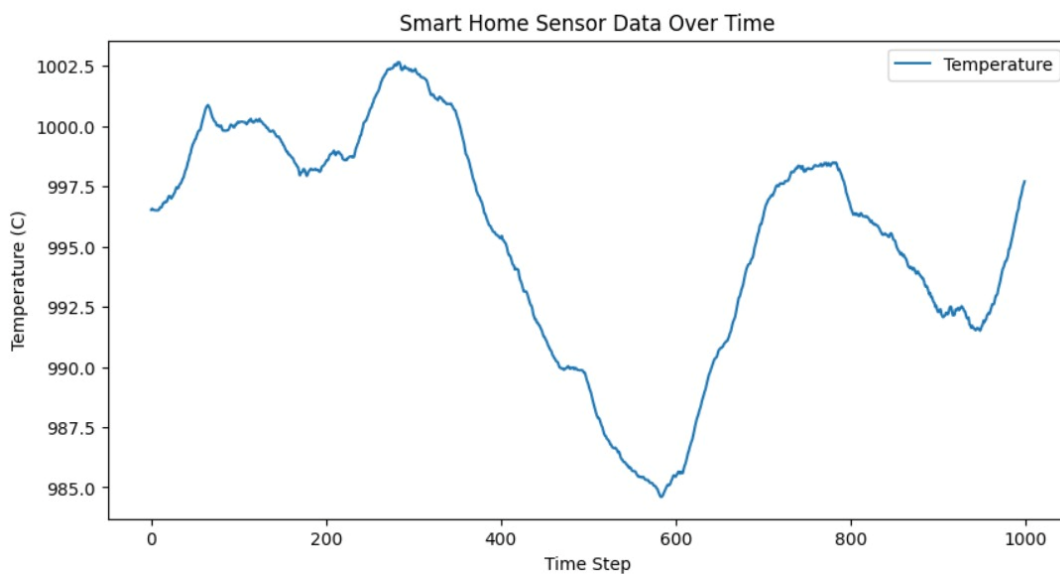
Modeling Approach

We treated this as a supervised regression problem.

- **Feature:** Current Temperature (T_t).
- **Target:** Future Temperature (T_{t+1}).
- **Model:** We trained a Linear Regression model to learn the relationship between the current state and the future state.

4. Results & Visualization

Dashboarding Below is the visualization of the temperature sensor data over time, demonstrating the cyclical nature of the dataset.



Model Performance Our Linear Regression model achieved the following performance on the test set:

- **Mean Absolute Error (MAE):** 0.0763 °C.

This result meets our success criteria defined in the problem statement.

5. Deployment

The project code is hosted on GitHub and configured for deployment on an AWS EC2 instance (Free Tier). The model script can be executed in a cloud environment to generate real-time predictions based on input data.